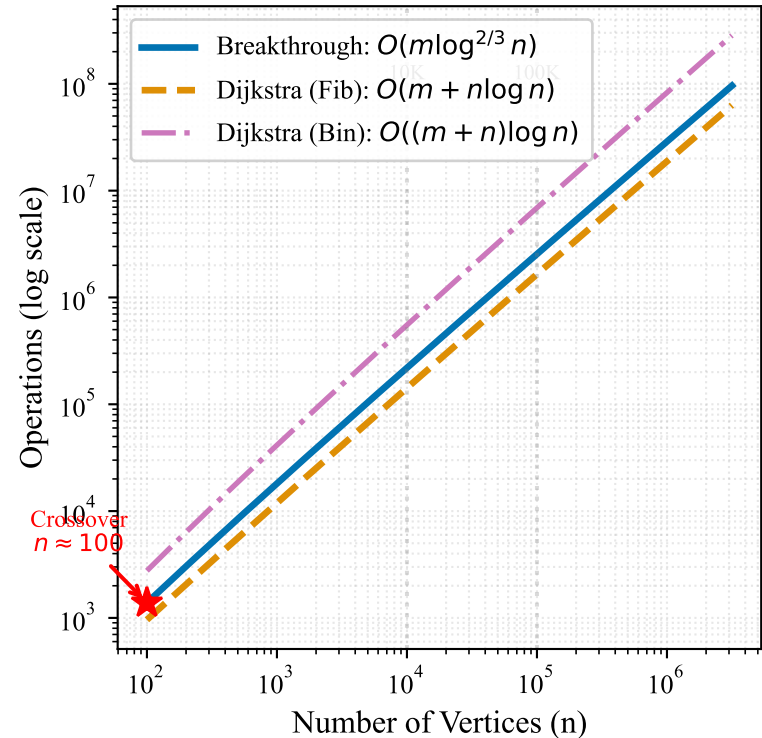
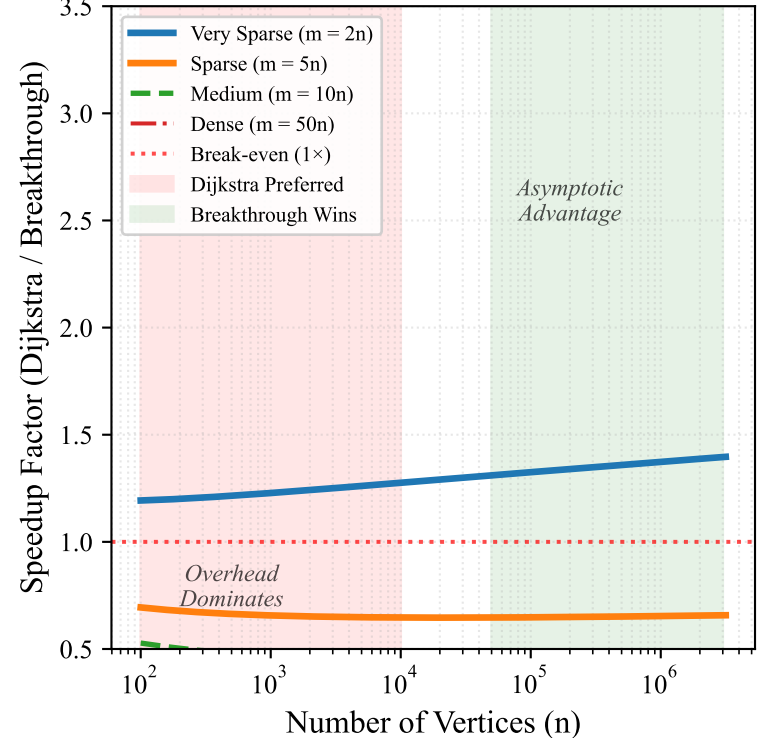


SSSP Algorithm Complexity Comparison: Breakthrough vs. Classical Algorithms

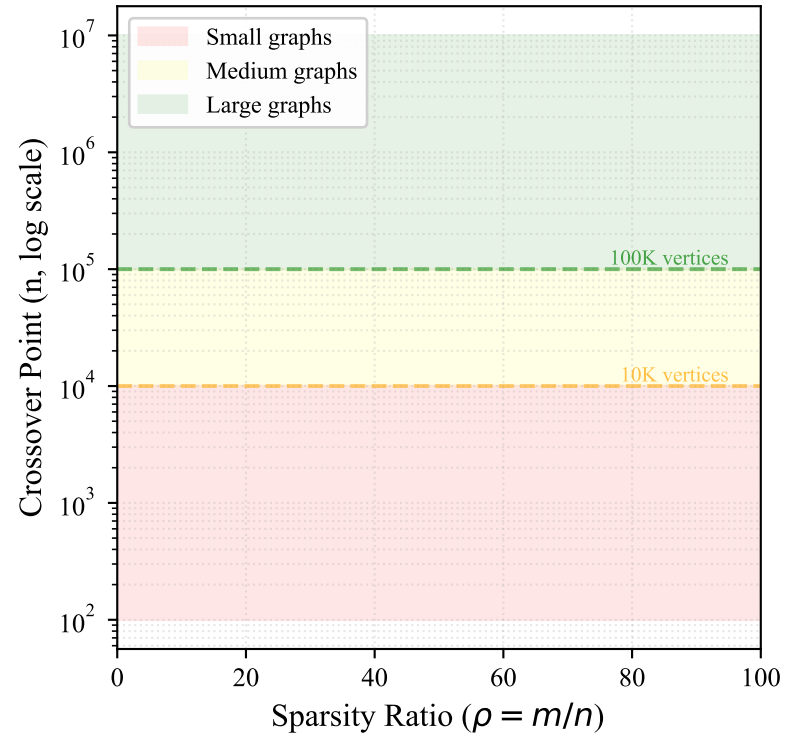
(a) Complexity Growth: Sparse Graphs ($m = 5n$)



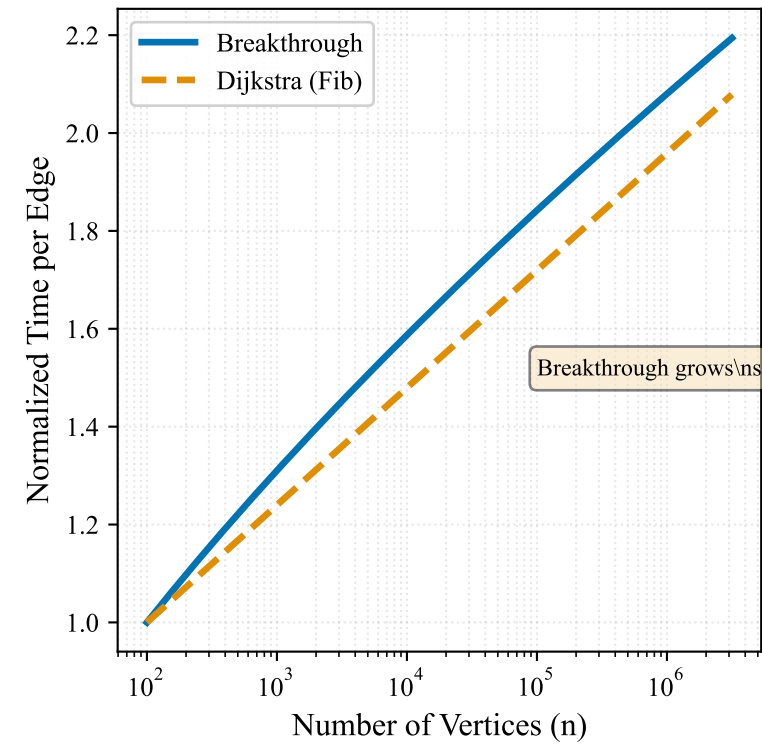
(b) Speedup Analysis: Varying Graph Densities



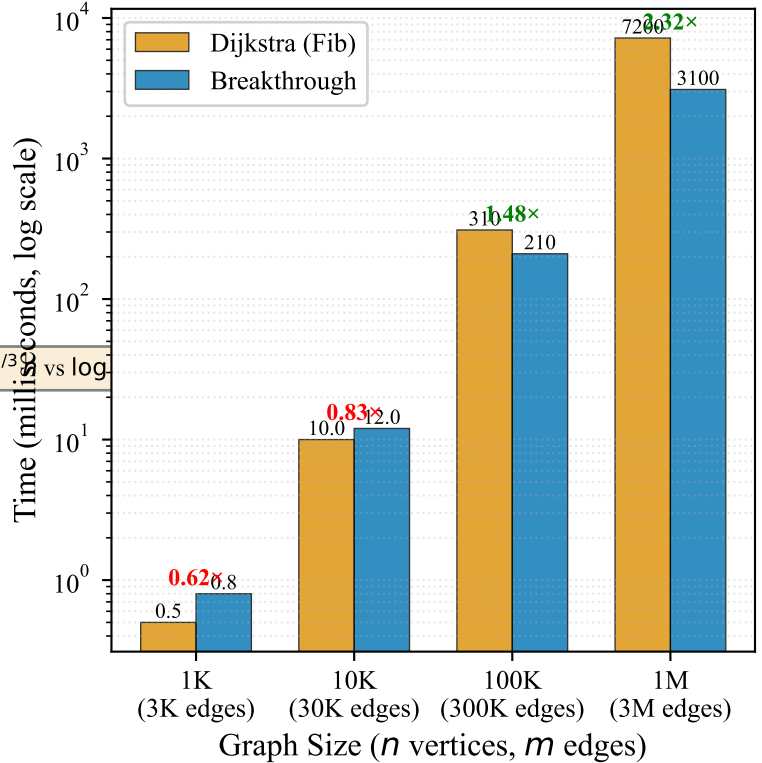
(c) Crossover Point vs. Graph Density



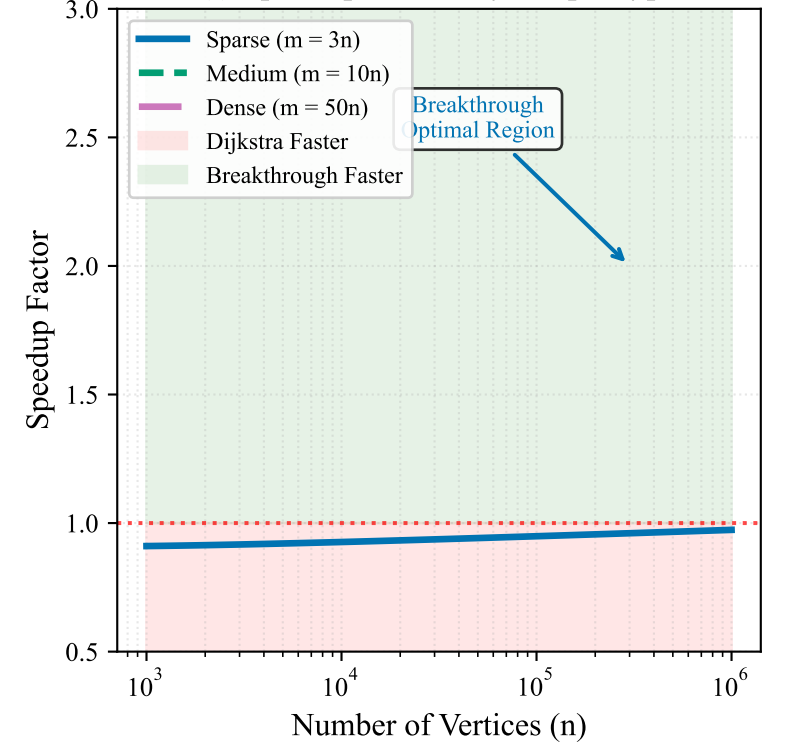
(d) Time per Edge Growth (Normalized to $n = 100$)



(e) Practical Performance Estimates



(f) Speedup Trends by Graph Type



Key Insights: (1) Breakthrough achieves $O(m \log^{2/3} n)$ vs Dijkstra's $O(m + n \log n)$ (2) Speedup increases with graph size (3) Optimal for sparse graphs with $n > 50,000$ vertices